My Notebook
Vacuum tech

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Ariel Smyrichinsky (arismyri@iu.edu) - Sep 21, 2026, 9:42 PM EDT

09.21.2026

8:43 PM Ariel Smyrichinsky

I have downloaded the Jupyter Notebook writeup for the Vacuum Technology lab.

It is called "P451_D_Vac_Tech_2026_Jun_09 (1).ipynb". Later, I plan on making my own notebook for the analysis.

The goal of this lab is to become acquainted with lab vacuum systems. In many experiments, vacuum systems are important for maintaining a space of little to no contaminants. A vacuum is defined as a region devoid of matter; this is helpful, because a space devoid of air means that there could be no conduction temperature loss of a sample from the air, and gaseous atoms won't be interfering with other results. Of course, a perfect vacuum is impossible to achieve, but getting close is helpful.

In this lab, we will compare theoretical predictions of the conductance of an opening to their actual values. Conductance is the constant of proportionality that turns a pressure differential into a gas flow.

Q = M/t = C_{12}(P_2 - P_1)

Where Q is the gas flow,

M is the amount of gas in Torr-liters (M = pV)

C_{12} is the conductance between volumes 1 and 2,

and P_1, P_2 are the pressures in volumes 1 and 2.

This lab utilizes a pumping station with various types of vacuum pumps, gauges, and valves:

Pumps: There are mechanical fore (force?) pumps (labeled FP 1 and 2.) They are rotating-vane pumps, which rotate in order to displace fluids.

Thermocouple gauges. Per this link on ScienceDirect: "A thermocouple gauge is a device that measures pressure based on the thermal conductivity of gases, utilizing a heated metal strip whose temperature change indicates the pressure level within a low-pressure range of 10 −4 mbar to 1 mbar."

A convection gauge. Per this link: "Convectron (sic) gauges take advantage of heat loss due to convection at higher pressures. This extends the range of accurate, repeatable measurement to atmosphere."

Comparing the thermocouple gauge operating mechanism to the convection gauge mechanism: Both indirectly measure pressure, but one uses conduction between a metal strip and the air and the other determines the loss of heat through fluids.

Bourdon gauge. This gauge directly measures pressure via mechanical deformation.

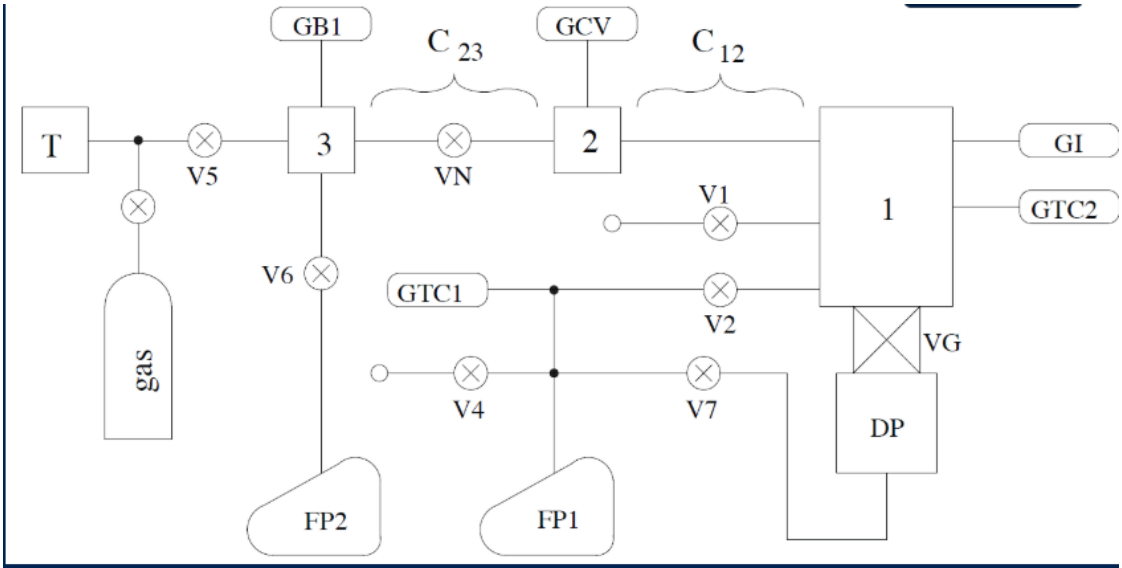


Fig. 1 Overall layout of the pumping station used in this lab. In order to observe the transition between viscous and molecular flow regimes of conductance for C12, you will need to measure the volume of V2 and install a known aperture betwen V1 and V2.

This image is taken directly from the lab writeup.

Ariel Smyrichinsky (arismyri@iu.edu) - Sep 21, 2026, 9:55 PM EDT

9:42 PM

Procedure for the lab is as follows (reworded from the writeup)

1. Close all the valves except the gate valve.

2. Turn on GCV (the convection gauge), GTC 1 and 2 (the thermocouple gauges), FP (fore? pump) 1 using the power strip on the back of the vacuum station.

3. Open V7 and V2 (valves) to evacuate volume 1. This helps to limit backflow of oil from the pump into the vacuum chamber.

4. Monitor GTC 1 and 2 and GCV as a function of time during the pump-down period. Do not turn on the ion gauge GI until later.

5. Wait until GTC 1 is stable, which is below 100 mTorr.

6. Turn on the diffusion pump cooling water.

7. Close V2 to isolate volume 1 from pump FP1. Leave V7 open.

8. Engage the [word omitted in writeup?] for the diffusion pump.

9. Wait 30 minutes, or until GTC 2 reads near zero and does not appear to be decreasing.

10. Turn on the ion gauge Gi and periodically record its values. The pressure in volume 1 will slowly decrease over the course of an hour and approach a base pressure of 10^-6 Torr.

Ariel Smyrichinsky (arismyri@iu.edu) - Oct 20, 2026, 8:03 PM EDT

09.22.2026

1:02 PM **Ariel Smyrichinsk**y, Garrett Willis, Camden Bush

We are inspecting the apparatus.

[Insert image of apparatus images matched to diagram.]

Willis has measured the orifice diameter, and doing so by finding a needle with varying width, and finding where the width of the needle matched the orifice diameter. A Mitutoyo micrometer was used to measure the needle width.

1.144 +/- 0.001 mm.



Next, we are measuring the length of the orifice channel with a dial caliper.

28.02 +/- 0.01 mm.

The goal is to measure the conductance of the orifice between volumes 1 and 2.

We have turned on the apparatus. Initially there were issues; the vacuum was not pumping, and therefore the Convectron gauge was not lowering, which was indicative to something afoot.

After trial and error, it was discerned that the issues were because V7 was not fully open. V7 is the valve between the pump and Volume 1, so V7 not fully open affected the pumping that would be occuring.

Right now, every valve is closed besides V7. The oil is in DP. It gets heated up, hits the walls within DP, and removes the air molecules from Volume 1 because DP is exposed to Volume 1 through VG.

2:13 PM

Right now, we are on step four and five of the startup process: monitoring GTC 1, GTC 2, GCV over time, and waiting for GTC 1 to be below 100 mTorr.

We have now turned on the ion gauge GI. Time marked in a Google Sheet Bush made and sent around.

The diffusion pump cooling water has been turned on for a while.

Now, GTC2 reads near zero and has not been visibly decreasing.

2:28 PM

Willis is manning the data collection station.

I have taken over the data collection station.

We are using a Windows desktop that has the program LoggerPro downloaded.

Connected to the desktop is a Vernier LabPro, with its channel 1 connected to the GCV.

This conveys the voltage of the GCV, which can act as a pressure reading.

When air is let into the vacuum through V6, the GCV voltage changes, and acts as a pressure reading.

We will be letting air into the vacuum and watching the GCV voltage change on LoggerPro.

On LoggerPro, we are taking data for 180 seconds each time, at ten samples per second.

2:48 PM

We are shutting off the ion gauge.

2:59 PM

We have completed our first run. Bush was able to get the diffusion pump to turn on.

I have saved data and a screenshot of this LoggerPro run as vacuum_run_1.txt and vacuum_run_1_screenshot.png.

Changed data collection time to 5 minutes (300 s).

Ariel Smyrichinsky (arismyri@iu.edu) - Sep 22, 2026, 4:36 PM EDT

3:36 PM

We have taken our last non-helium data.

We have now hooked up the helium gas.

3:47 PM

We are now about to take helium data.

4:10 PM

Have started the second helium run. The first and the second might be part of the same data set

Starting the third helium run.

4:36 PM

Commencing shut down process.

Ariel Smyrichinsky (arismyri@iu.edu) - Oct 21, 2026, 12:26 PM EDT

10.05.2026 6:20 PM

I am now going through the Jupyter Notebook with the intent of doing the analysis. The notebook is the same file as the writeup file.

The name of it when it was downloaded from Canvas was P451_D_Vac_Tech_2026_Jun_09 (1).ipynb, but I am renaming it to smyrichinsky_vacuum_lab.ipynb.

I am going to go through each block in order to make sure I understand it.

The first defined function, called pfromV_Air_Scalar, takes the voltage measurement passed to it and returns the value of the pressure. The docstring says that it gets these values used for each voltage from Table 3-6 in the Mini-Convectron Module Instruction Manual - 275512 - Rev. 03.

Depending on which value the voltage is below, the pressure will be found in different ways:

```
if V<0.375:
  raise Exception('Voltage too low V=%f'%V)
  return -99
if V<2.842:
  a = -0.02585
  b =  0.03767
  c =  0.04563
  d =  0.1151
  e = -0.04158
  f =  0.008737
  return a+V*(b+V*(c+V*(d+V*(e+V*f))))
elif V<4.945:
  a =  0.1031
  b = -0.3986
  c = -0.02322
  d =  0.07438
  e =  0.07229
  f = -0.006866
  return (a+V*(c+V*e))/(1.0+V*(b+V*(d+V*f)))
elif V<5.659:
  a = 100.624
  b = -0.37679
  c = -20.5623
  d = 0.0348656
  return (a+V*c)/(1.0+V*(b+V*d))
else:
  raise Exception("Invalid input voltage V=%f"%V)
return -99
```

Whichever value gets returned from the function above gets turned into a vector with np.vectorize.

The next defined function is pHefromPair, which, according to the docstring, "Implements the curves in figure 3-6 and 3.7 based on my piece-wise polynomial fits to ~3 decades of overlapping ranges of the log-log plots."

It takes the value P and scales it according depending on what value it is.

```
  x = math.log10(P)
  if x<-1.5:
    a = 0.22
    b = 1.39
    c = 0.521
    d = 0.372
    e = 0.144
    f = 0.0283
    g = 2.2E-03
    y = a+x*(b+x*(c+x*(d+x*(e+x*(f+x*g)))))
    return 10.0**y
  elif x<0.5:
    a = -0.0147
    b =  0.946
    c = -0.0958
    d = -0.271
    e = -0.07
    f =  0.101
    g =  0.0405
    y = a+x*(b+x*(c+x*(d+x*(e+x*(f+x*g)))))
    return 10.0**y
  else:
    a =  0.017
    b =  0.934
    c = -0.425
    d =  0.223
    e = -0.164
    f =  0.0615
    g = -7.78E-03
    y = a+x*(b+x*(c+x*(d+x*(e+x*(f+x*g)))))
    return 10.0**y
  raise Exception('Domain error.')
```

Whichever value is returned gets turned into a vector.

Then,. there are two testing functions: test_pfromV_Air and test_pHeFromPair, which takes the voltage V and pressure P respectively and plots them so they can be compared to existing tables.

The next section in the code is called "Compute theoretical values for the Conductance." Per the writeup, the conductance depends on properties of the tube, the pressure head between the two ends, and the gas properties. The flow could be molecular, viscous, or in a transitory state between the two.

The first defined function in this section is etaatT, which takes an eta value and a temperature T, then returns the viscosity of that temperature.

The next function, test_etaAtT tests that by comparing it with existing data.

The results get plotted.

The next function, called viscousC, takes the diameter of the pipe, the length of the pipe, the atomic mass of the gas, the temperature of the gas, the viscosity coefficient, and the pressure taken on both sides of the pipe and returns the conductance for viscous flow.

```python
import math
def viscousC(D, L, T, M, eta0, P1, P2):
    ''' Compute the conductance of a pipe assuming viscous flow

    Implements A. Roth, "Vacuum Technologies", Equation 3.53

    Parameters:
    -----------
    D : float
        Pipe diameter in cm
    L : float
        Pipe length in cm
    T : float
        gas temperature in kelvin
    M : float
        gas atomic mass
    eta0 : float
        coefficient of viscosity in poises at 270o (see eg. Roth, Table 2.9)
    P1 : float
        pressure on high side, torr
    P2 :
        pressure on low side, torr

    Note: Some of these are not used, but I wanted to keep the interface the same
    for the different conductance equations.

    Returns:
    --------
        float : the conductance in liters / second
    '''
    eta = etaAtT(eta0, T)
    Pbar = 0.5*(P1+P2)
    #
    # Roth gives the constant as 3.27E-2, but including the fudge factor 0.7 belo
    # gives a better match to external data (see tests below)
    #
    fac  = 0.70  # 3.27E-2
    return fac*(D**4/eta/L)*Pbar
```

The function molecularC does the same thing but for molecular flow.

```python
import math
def molecularC(D, L, T, M, eta0, P1, P2):
    ''' Compute the conductance of a pipe assuming molecular flow

    Implements A. Roth, "Vacuum Technologies", Equation 3.93

    Parameters:
    -----------
    D : float
        Pipe diameter in cm
    L : float
        Pipe length in cm
    T : float
        gas temperature in kelvin
    M : float
        gas atomic mass
    eta0 : float
        coefficient of viscosity in poises at 270o (see eg. Roth, Table 2.9)
    P1 : float
        pressure on high side, torr
    P2 :
        pressure on low side, torr

    Note: Some of these are not used, but I wanted to keep the interface the same
    for the different conductance equations.

    Returns:
    --------
        float : the conductance in liters / second
    '''
    return 3.81*math.sqrt(T/M)*D**3/L
```

The function test_flow_equations test and plots the data, and compares them to existing data from this link:

https://vacaero.com/information-resources/vacuum-pump-technology-education-and-training/9365-conductance-and-throughput-in-vacuum-

The next two sections are called "Low-level analysis" and "High-level analysis."

In "Low-level analysis," there are functions written in order to find the conductance as a function of pressure from the measured voltages.

The function analyze_chunk takes an array of time and a corresponding array of pressure, then finds the change in pressure over time.

```
def analyze_chunk(t, P):
  ''' Find the pressure, and slope of the pressure (dp/dt) for a data range

  Parameters:
  t - np.array of floats
    Time measurements in seconds
  P - np.array of floats
    Pressure measurements in torr

  Returns:
    (P0, sigP0, dPdt, sigdPdt)
    P0      - (float) pressure at center of chunck (torr)
    sigP0   - error on above
    dPdt    - (float) time derivative of pressure (torr/s)
    sigdPdt - (float) error on above

  Method:
    The data inside the pressure-vs-time data chunk is fit to an exponential
    which is approximately correct for nearly constant pressures.

    P(t) = P0 * exp( gamma * (t-t0))

    t0 is chosen at the center of the time values t. Then the pressure
    is given by P0 and the slope dPdt = gamma * P0
  '''
  if (len(t)<3):
    #
    # Can't analyze a chunk with fewer than 3 points
    #
    return (-1,-1,-1,-1)
  imid = int(len(t)/2)
  t0   = t[imid]
  tfit = t-t0
  from scipy.optimize import curve_fit
  def exp_func(x,A,gamma):
    return A*np.exp(gamma*x)
  guess = [P[imid], -1.0]
  popt, pcov = curve_fit(exp_func, tfit, P, p0=guess)
  perr = np.sqrt(np.diag(pcov))

  P0          = popt[0]
  sigma_P0    = perr[0]
  gamma       = popt[1]
  sigma_gamma = perr[1]
  dPdt        = gamma*P0
  sigma_dPdt  = dPdt*math.sqrt((perr[0]/popt[0])**2+(perr[1]/popt[1])**2)
  return (P0, sigma_P0, dPdt, sigma_dPdt)
```

There is the function setup_experiment, which takes the experiment values (size of volume 1, tube diameter, length of tube, and temperature of lab) and prints them.

There is the function analyze_conductance, which takes the voltage data, the name of the data, and the type of fluid inside, and finds the conductance.

The section called "High-level analysis":

Has a function called analyze_data_set, which takes the file name, the start and end times of the analysis, and the type of fluid and calls analyze_conductance on that data.

The last section is called "Top-level analysis," which is a singular code block that takes the system measurements (V1, d, L, and T) in the function setup_experiment, then runs analyze_data_set.

Ariel Smyrichinsky (arismyri@iu.edu) - Oct 21, 2026, 1:17 PM EDT

7:42 PM

I'm going to use the "Top-level analysis" code block and change the necessary variables to work with the data I have.

When trying to run the code, I found I had to alter the headers inside the data files so that the only column heads are t and v.

File names for data:

Air: vacuum_run_1.txt, vacuum_run_2.txt, vacuum_run_3.txt, vacuum_run_4.txt

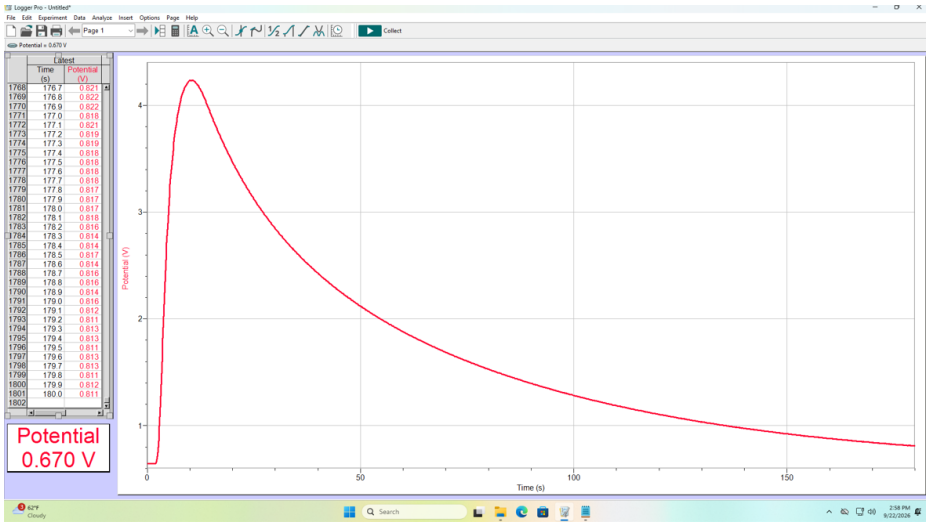
Helium: vacuum_run_1+2_He.txt, vacuum_run_3_He.txt (I plan on manually separating 1 and 2; this was a recording error in person. We forgot to start a new run on LoggerPro when taking data the second time.)

Also, when trying to run the code, I found that I had to specify that the data is tab-separated when reading in the data. I changed that line to pd.read_csv("[Name of data].txt", sep = "\t").

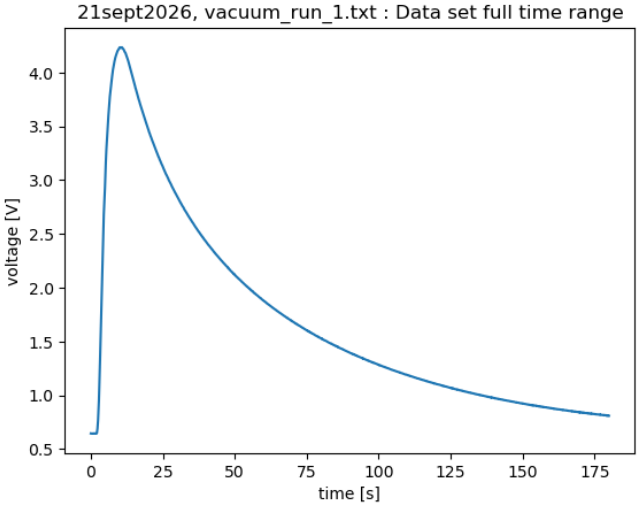
In the function viscousC, I implemented the constant being 3.27E-2 instead of 0.7.

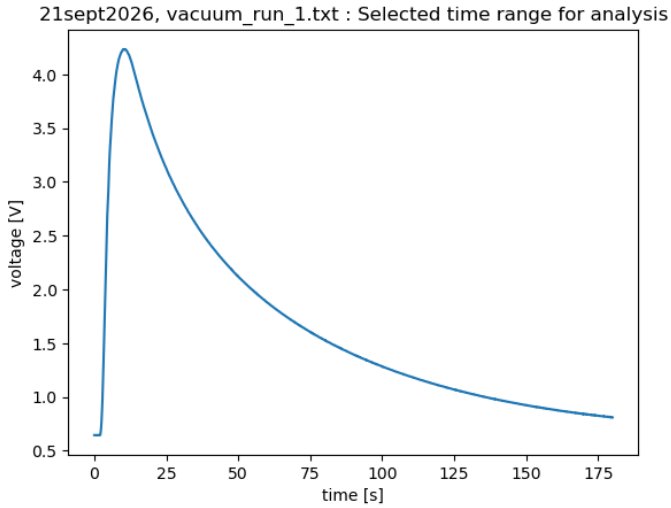
The first air run:

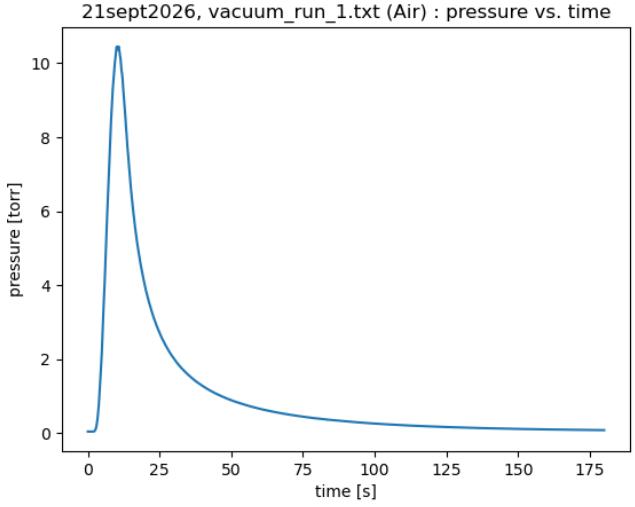
Logger pro screenshot:

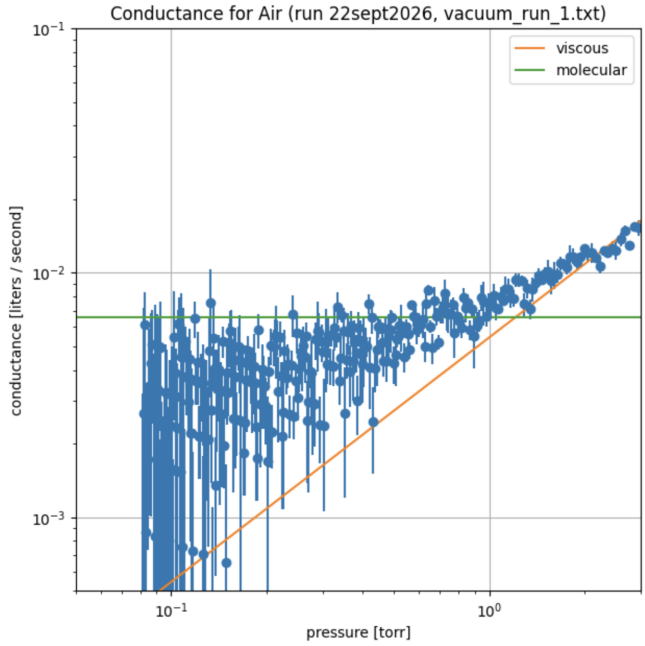


Graphs from analysis notebook: **** The dates should say 22Sept2026****

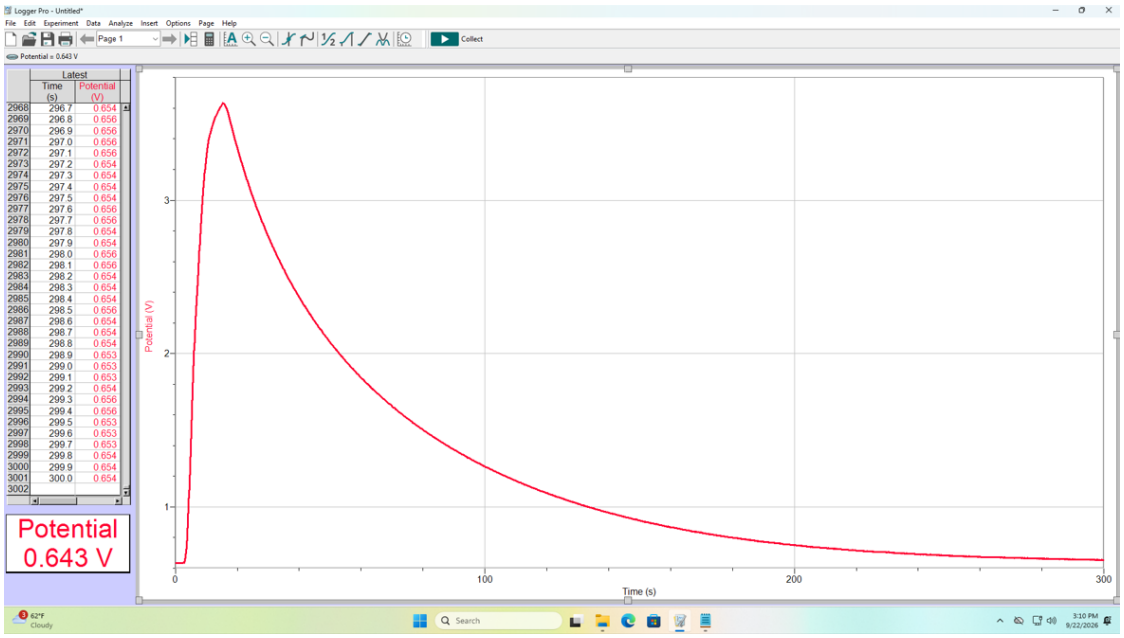

21sept2026, vacuum_run_1.txt : Data set full time range
voltage [V]
time [s]

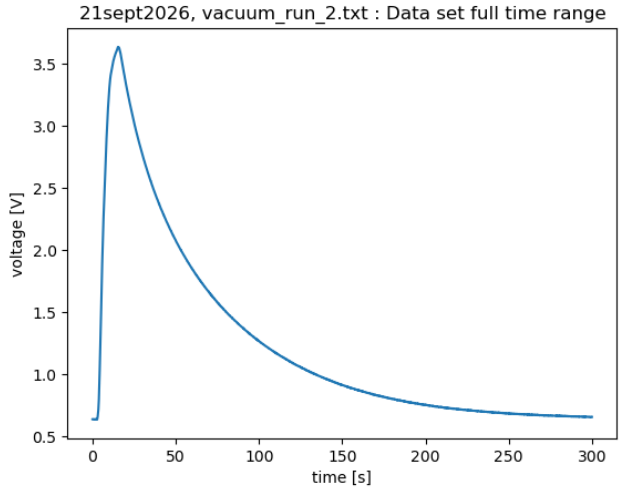

21sept2026, vacuum_run_1.txt : Selected time range for analysis
voltage [V]
time [s]

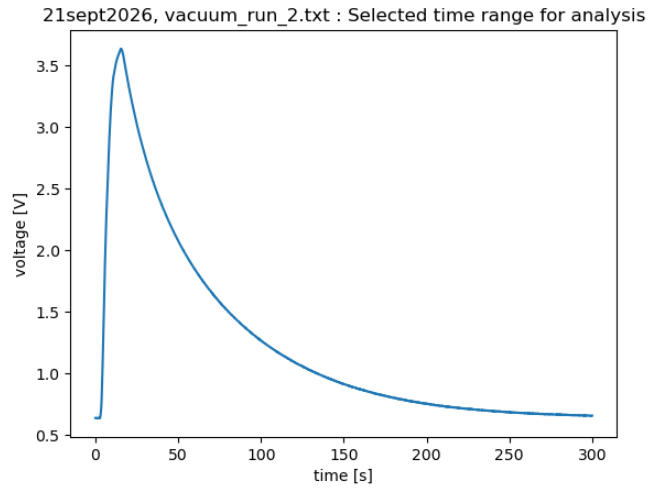

21sept2026, vacuum_run_1.txt (Air) : pressure vs. time
pressure [torr]
time [s]

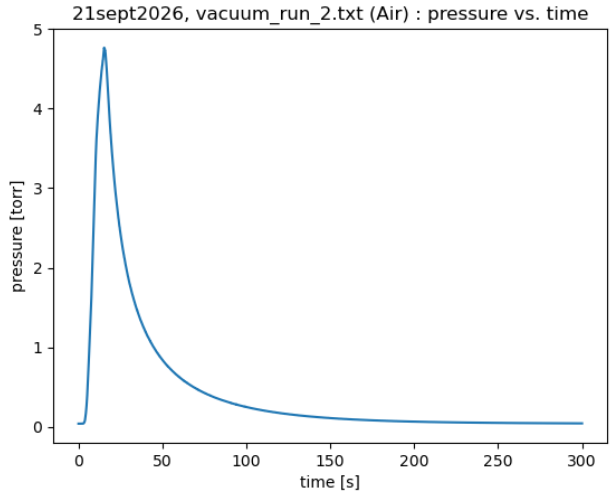


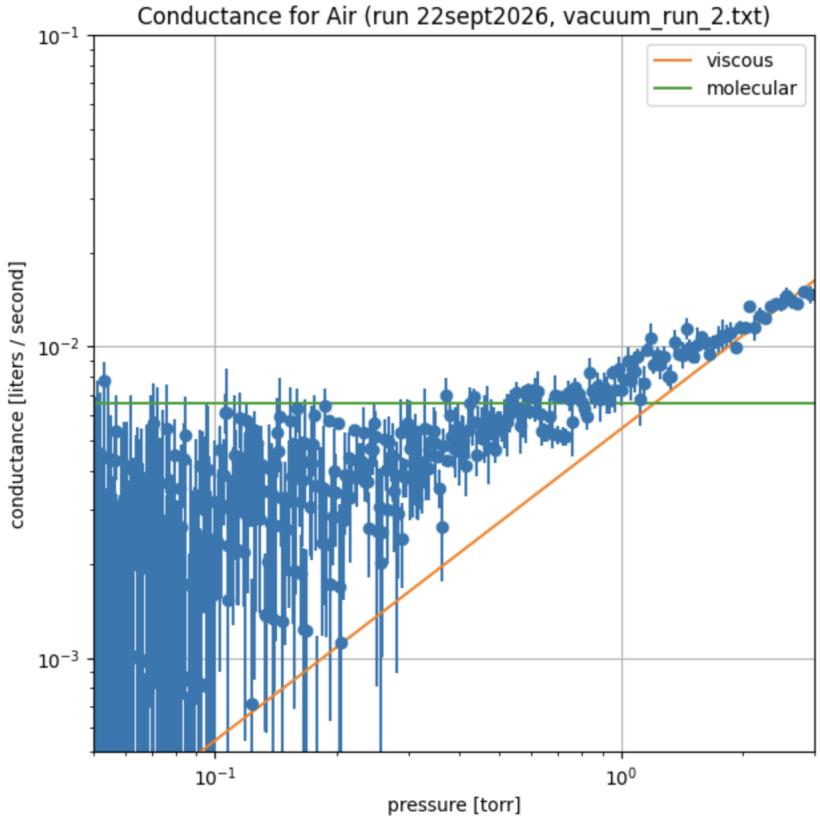
The second air run: **** The dates should say 22Sept2026****



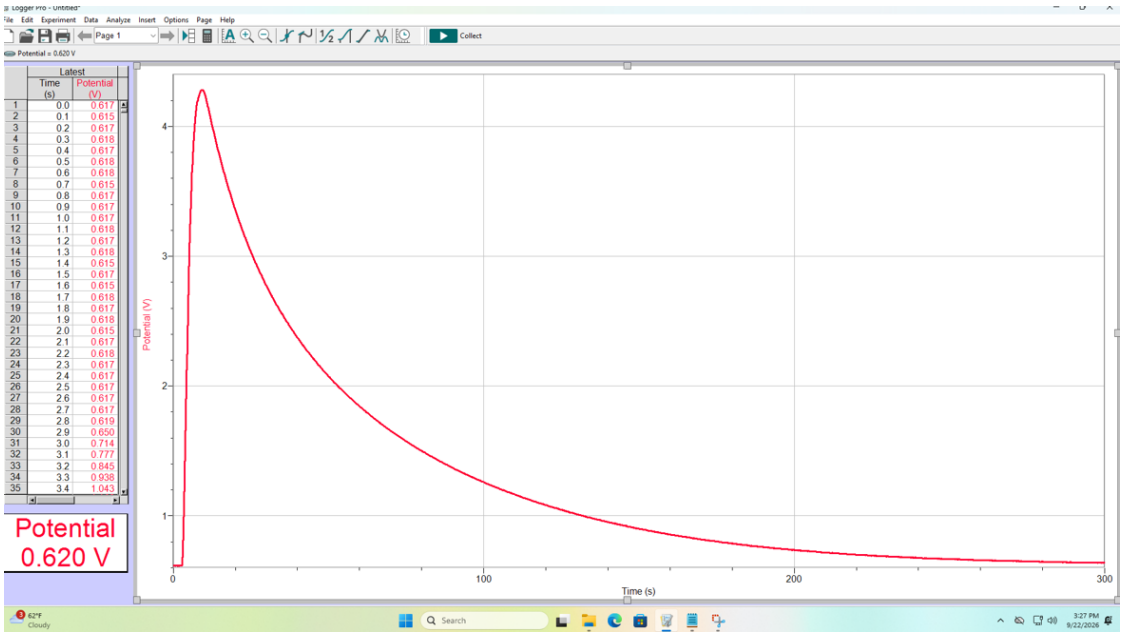

21sept2026, vacuum_run_2.txt : Data set full time range
voltage [V]
time [s]

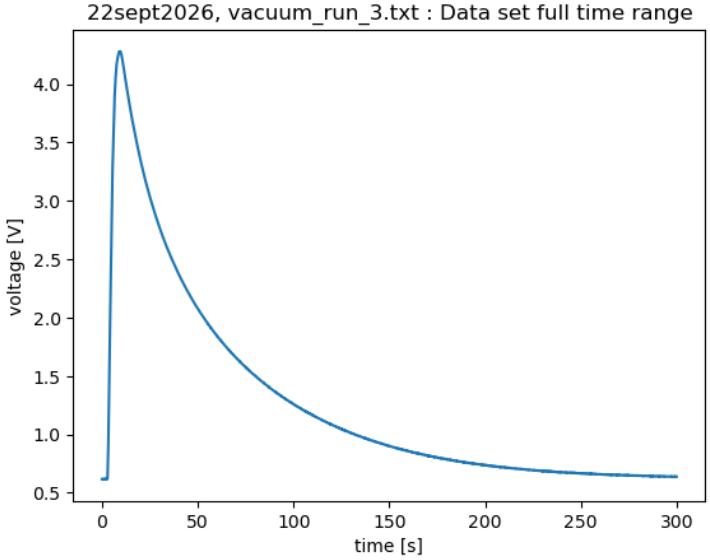

21sept2026, vacuum_run_2.txt : Selected time range for analysis
voltage [V]
time [s]

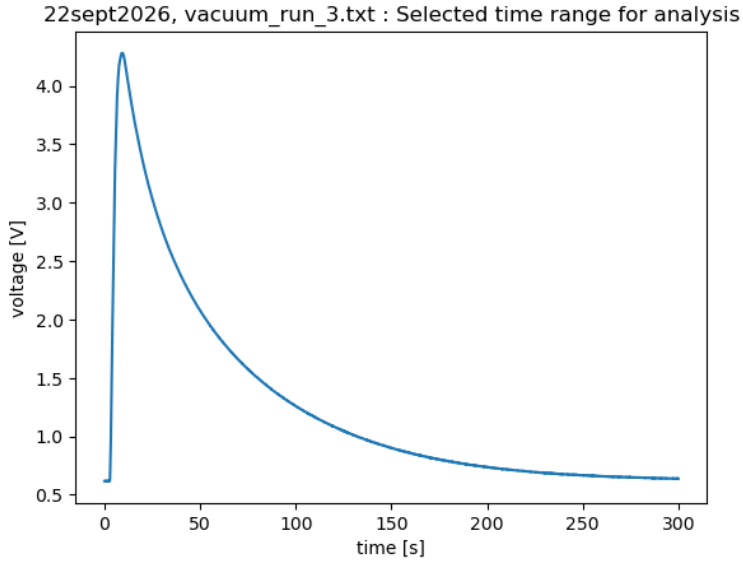

21sept2026, vacuum_run_2.txt (Air) : pressure vs. time
pressure [torr]
time [s]

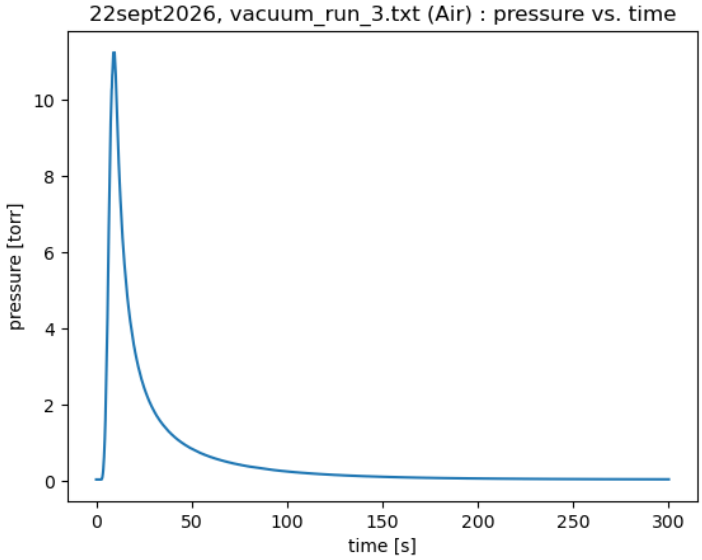


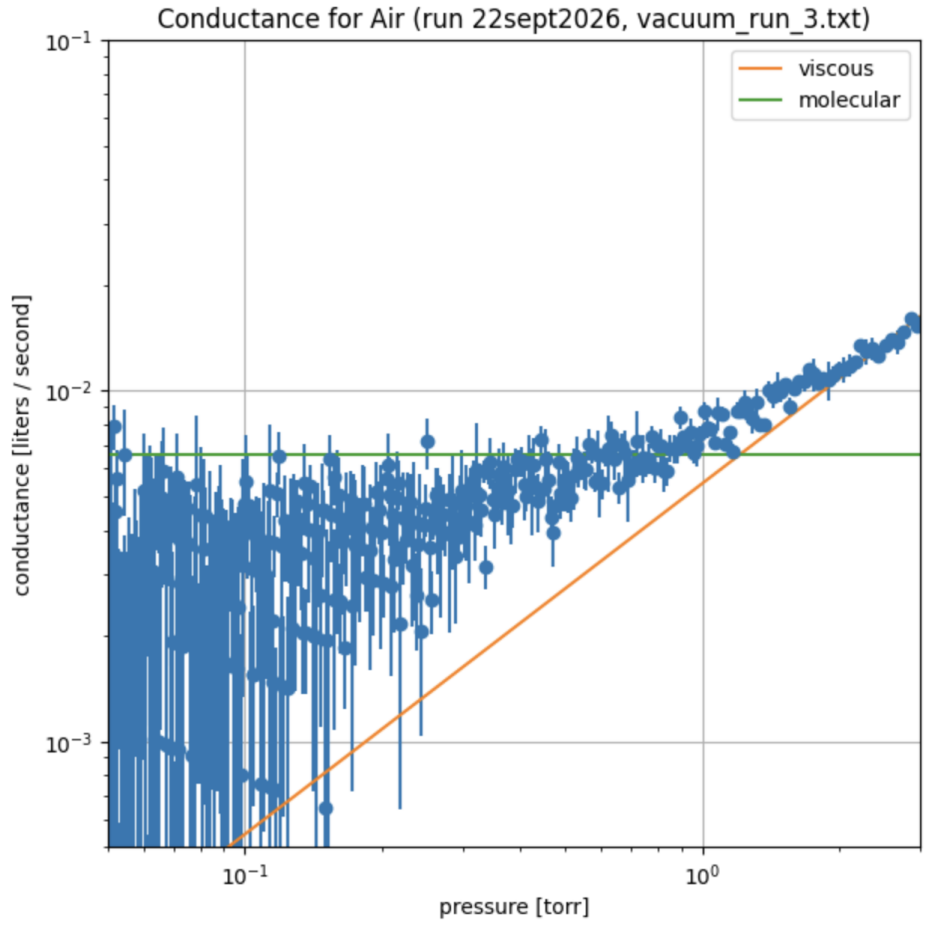
Third air run:



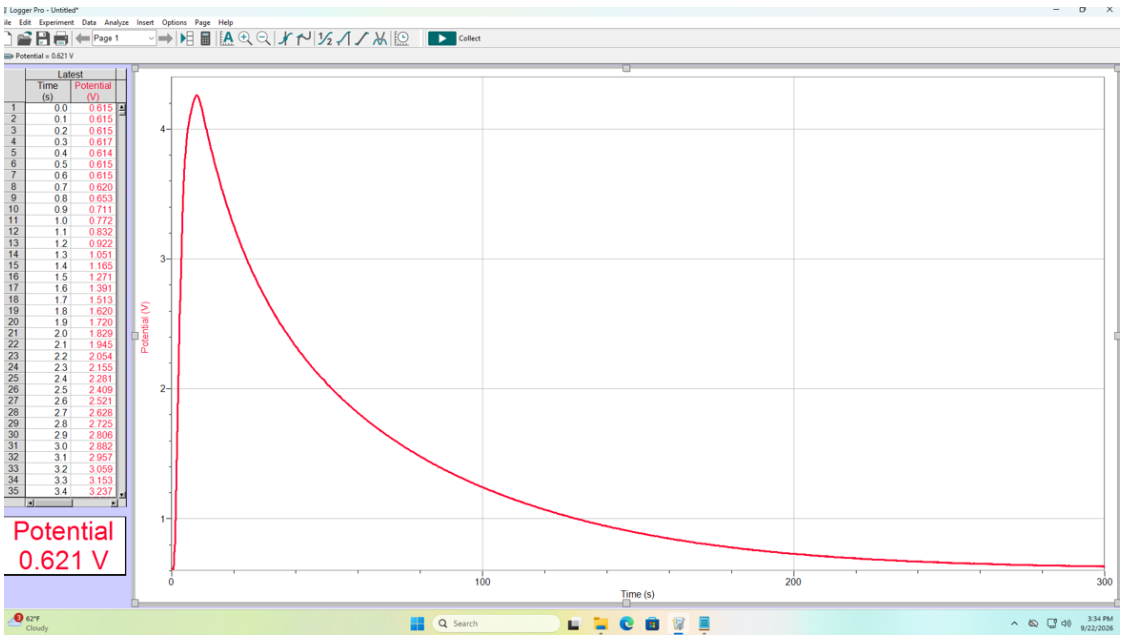

22sept2026, vacuum_run_3.txt : Data set full time range
voltage [V]
time [s]
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
50
100
150
200
250
300

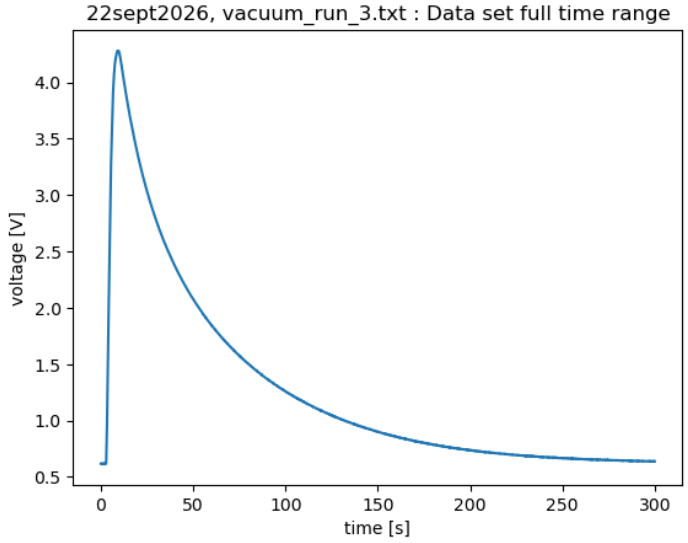

22sept2026, vacuum_run_3.txt : Selected time range for analysis
voltage [V]
time [s]
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
50
100
150
200
250
300

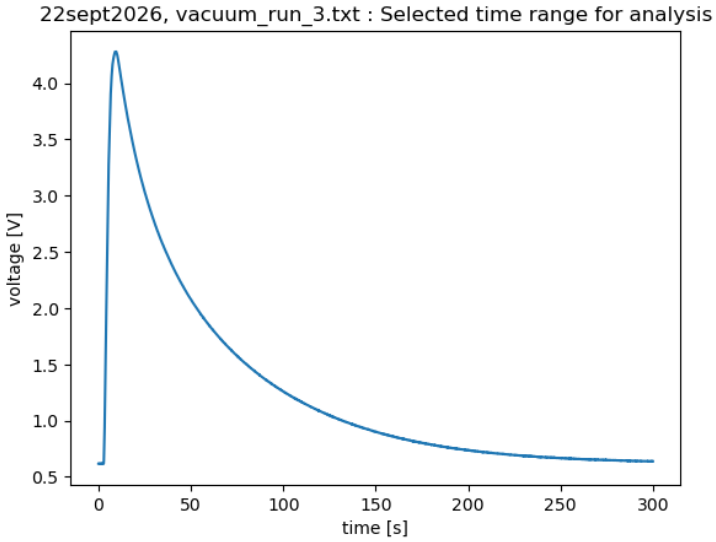

22sept2026, vacuum_run_3.txt (Air) : pressure vs. time
pressure [torr]
time [s]
0
2
4
6
8
10
0
50
100
150
200
250
300

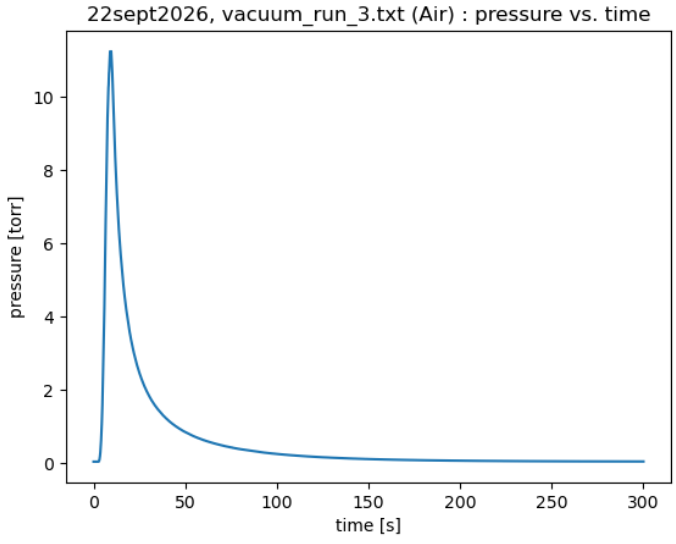


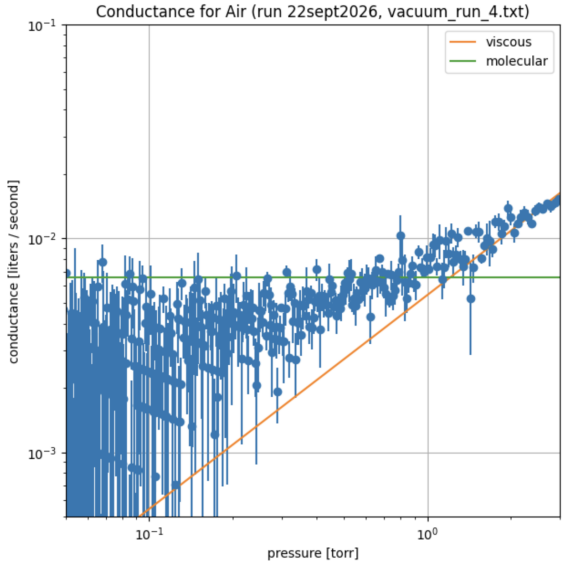
Fourth air run:




22sept2026, vacuum_run_3.txt : Data set full time range
voltage [V]
time [s]
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
50
100
150
200
250
300


22sept2026, vacuum_run_3.txt : Selected time range for analysis
voltage [V]
time [s]
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
50
100
150
200
250
300


22sept2026, vacuum_run_3.txt (Air) : pressure vs. time
pressure [torr]
time [s]
0
2
4
6
8
10
0
50
100
150
200
250
300

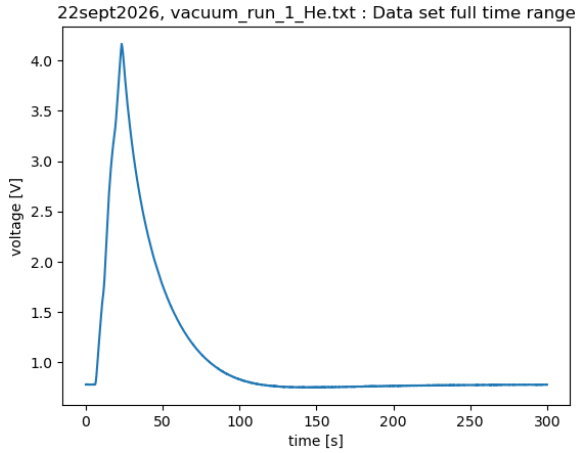


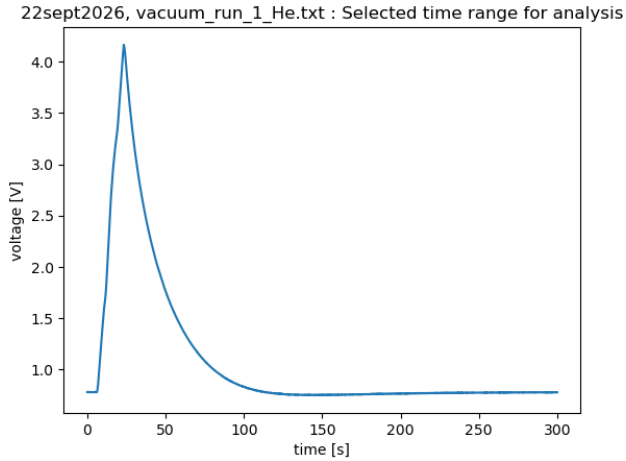
First helium run. When splitting vacuum_run_1+2_He.txt, I split them when t = 300, as we were doing runs for 300 seconds each. The files are now vacuum_run_1_He.txt and vacuum_run_2_He.txt.

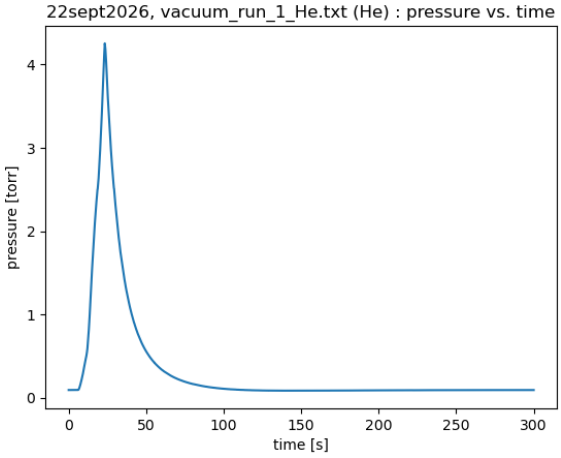
Ariel Smyrichinsky (arismyri@iu.edu) - Oct 21, 2026, 1:10 PM EDT

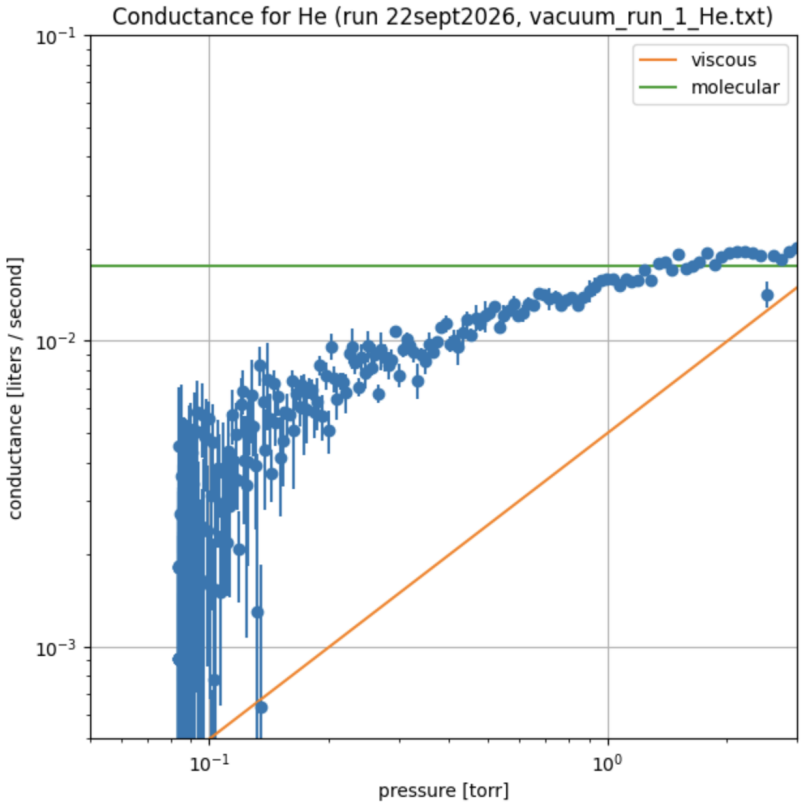
10.20.2026

Now actually pasting the actual first helium run results:

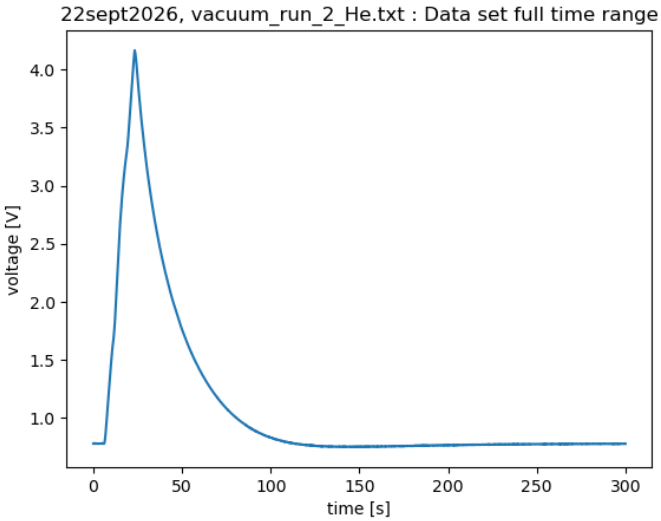

22sept2026, vacuum_run_1_He.txt : Data set full time range
voltage [V]
time [s]

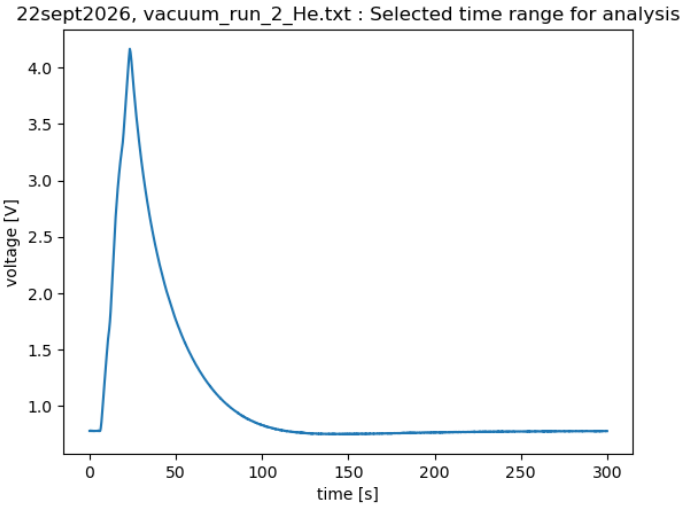

22sept2026, vacuum_run_1_He.txt : Selected time range for analysis
voltage [V]
time [s]

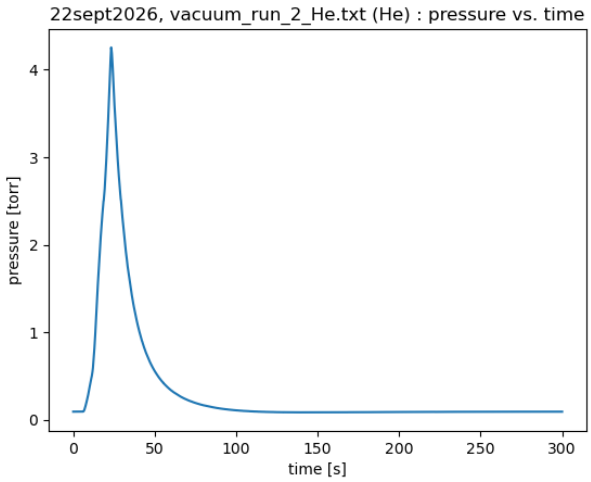

22sept2026, vacuum_run_1_He.txt (He) : pressure vs. time
pressure [torr]
time [s]

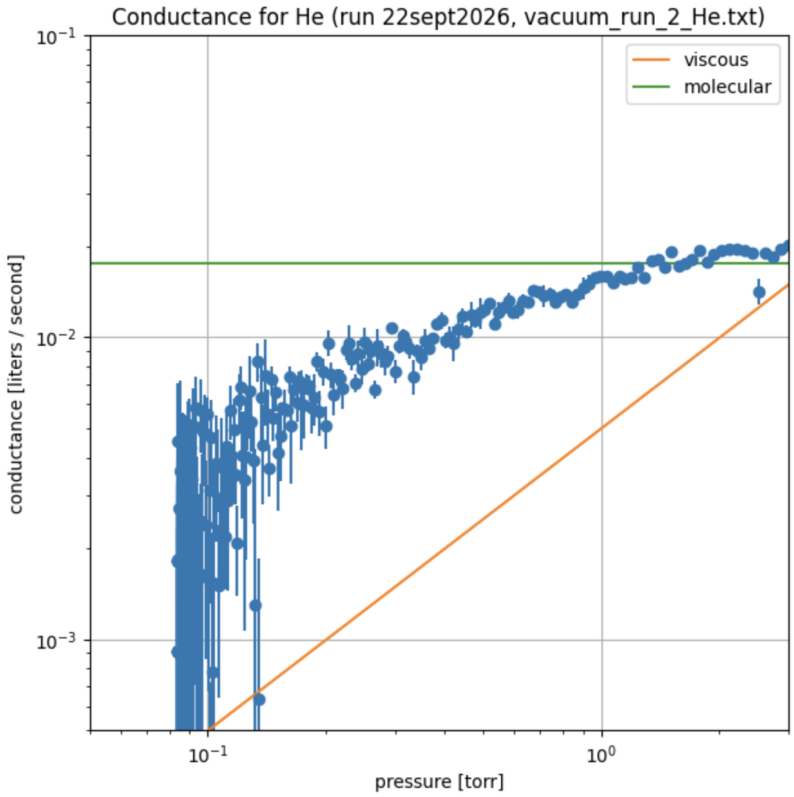


Second helium run:

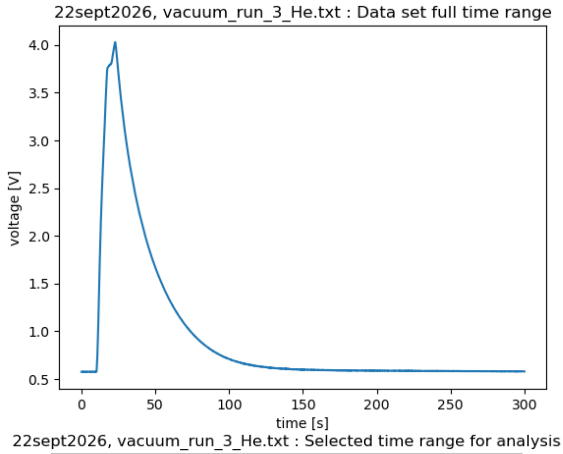


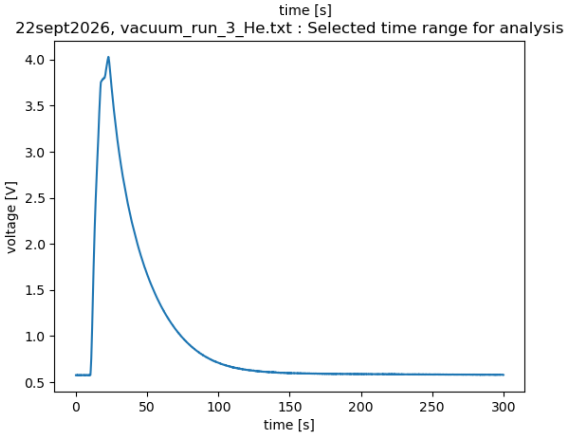


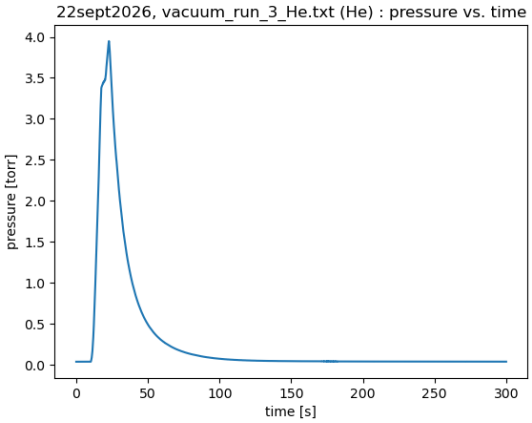


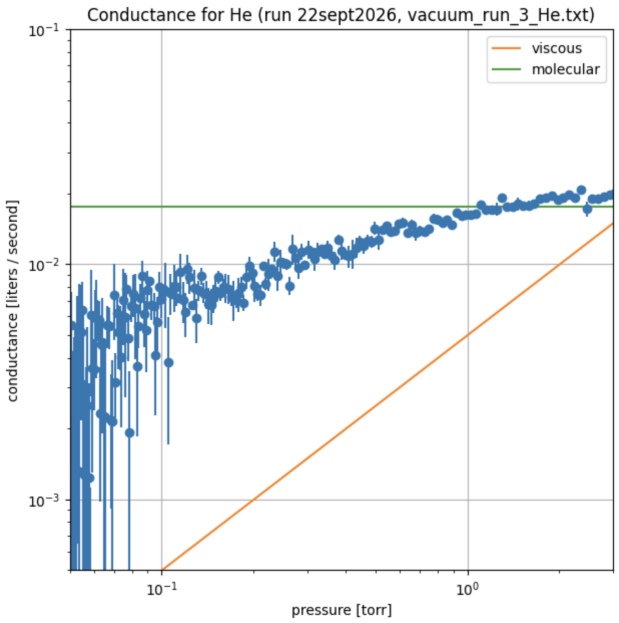


Third vacuum run:









Ariel Smyrichinsky (arismyri@iu.edu) - Oct 21, 2026, 2:38 PM EDT

10.21.2026

Conclusion: It is expected that as the pressure changes, the conductance will change as well. On the above graphs, there is a green line and an orange line. These lines represent the theoretical predictions what the conductance of the orifice between the two volumes will be for molecular and viscous flow respectively. The change in conductance is observed to be the case in all the runs with air and helium, however, it is most obvious in the runs that use air that the conductance goes from being constant at low pressure to being reliant on pressure. This could be the cause because helium is lighter than air, and is able to go to enter the molecular flow state faster than air.

The error bars are likely larger at low pressure because the gauges are more sensitive to fluctuations at those levels.